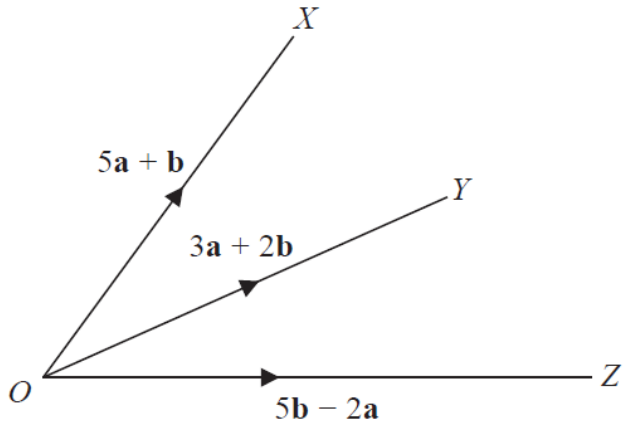
**21** The diagram shows the lines $OX$, $OY$ and $OZ$.




$\overrightarrow{OX} = 5\mathbf{a} + \mathbf{b}$

$\overrightarrow{OY} = 3\mathbf{a} + 2\mathbf{b}$

$\overrightarrow{OZ} = 5\mathbf{b} - 2\mathbf{a}$

Determine whether or not $XYZ$ is a straight line.
You must show how you get your answer.

$XY = OY - OX = (3a + 2b) - (5a + b)$

$XY = -2a + b$

**1 mark**
One difference vector.

$YZ = OZ - OY = (-2a + 5b) - (3a + 2b)$

$YZ = -5a + 3b$

**1 mark**
Two correct vectors.

*For multiples, a would require factor 2.5,*

*but b would require factor 3. These differ.*

*No: XY and YZ are not parallel, so XYZ is not straight.*

**1 mark**
Explain why the vectors are not multiples.

**(Total for Question 21 is 3 marks)**